

Internetworking

1. What is Internetworking?

- A) Connecting a computer to a printer
- B) Connecting multiple networks together to form a larger network
- C) Connecting a mouse to a keyboard
- D) Installing an operating system

Answer: B) Connecting multiple networks together to form a larger network

2. Which device is primarily used to connect different networks (Internetworking)?

- A) Hub
- B) Switch
- C) Router
- D) Repeater

Answer: C) Router

3. A collection of interconnected networks is called an:

- A) Internetwork
- B) Intranetwork
- C) Ultranetwork
- D) Extranet

Answer: A) Internetwork

4. Which device operates at the Network Layer of the OSI model?

- A) Bridge
- B) Repeater
- C) Router
- D) Gateway

Answer: C) Router

5. Which device is used to connect networks using different protocols?

- A) Router
- B) Bridge
- C) Gateway
- D) Switch

Answer: C) Gateway

6. Fragmentation is the responsibility of which layer?

- A) Data Link Layer
- B) Network Layer (Internetworking)
- C) Transport Layer
- D) Session Layer

Answer: B) Network Layer (Internetworking)

7. What is MTU?

- A) Maximum Transfer Utility
- B) Maximum Transmission Unit
- C) Minimum Transmission Unit
- D) Media Transfer Unit

Answer: B) Maximum Transmission Unit

8. When a packet is larger than the MTU of the network it is entering, what happens?

- A) It is dropped
- B) It is compressed
- C) It is fragmented
- D) It is returned

Answer: C) It is fragmented

9. Which protocol handles error reporting and query messages in the IP network?

- A) TCP
- B) ICMP
- C) UDP

D) ARP

Answer: B) ICMP

10. The process of reassembling fragmented packets usually happens at the:

A) Next router

B) Source host

C) Destination host

D) Intermediate gateway

Answer: C) Destination host

11. Which protocol is used to find the hardware address of a local device when its IP address is known?

A) RARP

B) ICMP

C) IP

D) ARP

Answer: D) ARP

12. A "Default Gateway" is essentially:

A) The IP address of the local router

B) The IP address of the DNS server

C) The MAC address of the switch

D) The Loopback address

Answer: A) The IP address of the local router

13. Which routing method uses a fixed routing table entered manually?

A) Dynamic Routing

B) Static Routing

C) Adaptive Routing

D) Default Routing

Answer: B) Static Routing

14. OSPF stands for:

- A) Open Shortest Path First
- B) Open Source Path First
- C) Open System Path First
- D) Only Shortest Path First

Answer: A) Open Shortest Path First

15. RIP (Routing Information Protocol) uses which metric to determine the best path?

- A) Bandwidth
- B) Hop Count
- C) Delay
- D) Load

Answer: B) Hop Count

16. What is the maximum hop count allowed in RIP?

- A) 10
- B) 15
- C) 16
- D) 255

Answer: B) 15 (16 is considered unreachable)

17. Which protocol is an Exterior Gateway Protocol (EGP) used between Autonomous Systems?

- A) RIP
- B) OSPF
- C) BGP
- D) IGRP

Answer: C) BGP (Border Gateway Protocol)

18. An Autonomous System (AS) is:

- A) A single computer

- B) A collection of networks under a single administrative domain
- C) The entire Internet
- D) A type of router

Answer: B) A collection of networks under a single administrative domain

19. NAT stands for:

- A) Network Access Translation
- B) Network Address Translation
- C) New Address Translation
- D) Node Access Tool

Answer: B) Network Address Translation

20. What is the primary purpose of NAT?

- A) To speed up routing
- B) To conserve public IP addresses
- C) To encrypt data
- D) To assign MAC addresses

Answer: B) To conserve public IP addresses

21. Tunneling is a technique used to:

- A) Dig cables underground
- B) Encapsulate one protocol inside another
- C) Connect two hubs
- D) Speed up Wi-Fi

Answer: B) Encapsulate one protocol inside another

22. Which IP address indicates a broadcast to the local network?

- A) 0.0.0.0
- B) 127.0.0.1
- C) 255.255.255.255
- D) 192.168.1.1

Answer: C) 255.255.255.255

23. CIDR stands for:

- A) Classless Inter-Domain Routing
- B) Classful Internet Domain Routing
- C) Computer Internet Domain Routing
- D) Core Inter-Domain Routing

Answer: A) Classless Inter-Domain Routing

24. In CIDR notation /24 represents a subnet mask of:

- A) 255.0.0.0
- B) 255.255.0.0
- C) 255.255.255.0
- D) 255.255.255.255

Answer: C) 255.255.255.0

25. The process of moving a packet from one network to another is called:

- A) Switching
- B) Forwarding / Routing
- C) Bridging
- D) Framing

Answer: B) Forwarding / Routing

26. Which layer does a Bridge operate at?

- A) Physical
- B) Data Link
- C) Network
- D) Transport

Answer: B) Data Link

27. A device that connects two LAN segments but blocks broadcast traffic is a:

- A) Hub
- B) Repeater
- C) Router
- D) Bridge

Answer: C) Router

28. Distance Vector routing algorithms suffer from which problem?

- A) Count-to-Infinity
- B) Shortest Path
- C) Flooding
- D) Deadlock

Answer: A) Count-to-Infinity

29. Which technique is used to solve the Count-to-Infinity problem?

- A) Tunneling
- B) Split Horizon
- C) Fragmentation
- D) Checksum

Answer: B) Split Horizon

30. Link State routing protocols flood the network with:

- A) Routing Tables
- B) Link State Advertisements (LSAs)
- C) Hello packets only
- D) Error messages

Answer: B) Link State Advertisements (LSAs)

31. Which of the following is a Link State protocol?

- A) RIP
- B) OSPF
- C) IGRP

D) BGP (Path Vector)

Answer: B) OSPF

32. Which mechanism allows multiple private IP addresses to map to a single public IP address?

A) Static NAT

B) PAT (Port Address Translation) / NAT Overload

C) Bridging

D) Switching

Answer: B) PAT (Port Address Translation) / NAT Overload

33. What is the loopback address used for?

A) Testing the network card and software stack of the local machine

B) Broadcasting

C) Identifying the router

D) Identifying the server

Answer: A) Testing the network card and software stack of the local machine

34. A specialized computer that connects two or more networks is called a:

A) Server

B) Workstation

C) Router

D) Client

Answer: C) Router

35. Which field in the IP header is decremented by each router?

A) TOS

B) TTL

C) Length

D) Checksum

Answer: B) TTL

36. When TTL reaches 0, the router:

- A) Forwards the packet
- B) Discards the packet and sends an ICMP Time Exceeded message
- C) Resets the TTL
- D) Returns the packet

Answer: B) Discards the packet and sends an ICMP Time Exceeded message

37. Which protocol provides logical addressing?

- A) Ethernet
- B) IP
- C) TCP
- D) PPP

Answer: B) IP

38. The method of dividing a network into smaller networks is called:

- A) Supernetting
- B) Subnetting
- C) Tunneling
- D) Bridging

Answer: B) Subnetting

39. The method of combining multiple small networks into a larger network route is called:

- A) Subnetting
- B) Supernetting (Route Aggregation)
- C) Fragmenting
- D) Dividing

Answer: B) Supernetting (Route Aggregation)

40. Which network class supports the largest number of networks?

- A) Class A
- B) Class B

C) Class C

D) Class D

Answer: C) Class C (Supports most networks, fewest hosts per network)

41. Which command identifies the routers used to reach a destination?

A) ping

B) tracert / traceroute

C) ipconfig

D) netstat

Answer: B) tracert / traceroute

42. In a router, the Routing Table contains:

A) MAC addresses

B) Destination IP networks and the next hop/interface

C) User passwords

D) File names

Answer: B) Destination IP networks and the next hop/interface

43. Which protocol resolves an IP address to a MAC address?

A) RARP

B) DNS

C) ARP

D) DHCP

Answer: C) ARP

44. Which protocol is used by routers to communicate availability to other routers?

A) Routing Protocols (RIP, OSPF, etc.)

B) ARP

C) DHCP

D) DNS

Answer: A) Routing Protocols

45. The "Internet" is an example of a:

- A) LAN
- B) WAN
- C) Global Internetwork
- D) PAN

Answer: C) Global Internetwork

46. Bridges filter traffic based on:

- A) IP Address
- B) MAC Address
- C) Port Number
- D) Protocol

Answer: B) MAC Address

47. Routers filter traffic based on:

- A) IP Address
- B) MAC Address
- C) Physical location
- D) Computer Name

Answer: A) IP Address

48. A brouter acts as:

- A) A Bridge and a Router
- B) A Bridge and a Repeater
- C) A Router and a Hub
- D) A Switch and a Hub

Answer: A) A Bridge and a Router

49. Which ICMP message is sent if a destination network is unreachable?

- A) Source Quench

- B) Destination Unreachable
- C) Time Exceeded
- D) Redirect

Answer: B) Destination Unreachable

50. IGMP is used for:

- A) Error reporting
- B) Group management in Multicast
- C) Routing
- D) Addressing

Answer: B) Group management in Multicast

51. IPv6 addresses are expressed in:

- A) Decimal
- B) Hexadecimal
- C) Binary
- D) Octal

Answer: B) Hexadecimal

52. The size of the IPv6 address space is:

- A) 2^{32}
- B) 2^{64}
- C) 2^{128}
- D) 2^{256}

Answer: C) 2^{128}

53. Which feature allows IPv6 to automatically configure an address?

- A) SLAAC (Stateless Address Autoconfiguration)
- B) NAT
- C) ARP
- D) OSPF

Answer: A) SLAAC

54. Dual Stack refers to:

- A) Running IP and IPX
- B) Running IPv4 and IPv6 simultaneously
- C) Running TCP and UDP
- D) Having two routers

Answer: B) Running IPv4 and IPv6 simultaneously

55. Which header field handles Quality of Service (QoS) in IPv4?

- A) TTL
- B) Type of Service (ToS) / DiffServ
- C) Version
- D) Identification

Answer: B) Type of Service (ToS) / DiffServ

56. VLAN stands for:

- A) Virtual Local Area Network
- B) Visual Local Area Network
- C) Virtual Large Network
- D) Vertical Local Area Network

Answer: A) Virtual Local Area Network

57. VLANs are configured on:

- A) Hubs
- B) Repeaters
- C) Switches
- D) Cables

Answer: C) Switches

58. Which standard describes VLAN tagging?

- A) IEEE 802.3
- B) IEEE 802.11
- C) IEEE 802.1Q
- D) IEEE 802.5

Answer: C) IEEE 802.1Q

59. Inter-VLAN routing requires a:

- A) Hub
- B) Layer 2 Switch
- C) Router or Layer 3 Switch
- D) Repeater

Answer: C) Router or Layer 3 Switch

60. What is a "Hop" in internetworking?

- A) A data collision
- B) A connection error
- C) The passage of a packet through a router
- D) A physical cable

Answer: C) The passage of a packet through a router

61. Proxy ARP allows:

- A) A router to answer ARP requests on behalf of a remote host
- B) A switch to route
- C) A PC to act as a server
- D) A faster ping

Answer: A) A router to answer ARP requests on behalf of a remote host

62. DHCP Relay Agent is needed when:

- A) The DHCP server is on the same subnet
- B) The DHCP server is on a different subnet
- C) There is no DHCP server

D) Static IPs are used

Answer: B) The DHCP server is on a different subnet

63. Which metric does OSPF use?

A) Hop Count

B) Cost (Bandwidth)

C) Reliability

D) Load

Answer: B) Cost (Bandwidth)

64. Area 0 in OSPF is called the:

A) Stub Area

B) Backbone Area

C) Totally Stubby Area

D) Local Area

Answer: B) Backbone Area

65. EIGRP is a:

A) Link State Protocol

B) Distance Vector Protocol

C) Hybrid (Advanced Distance Vector) Protocol

D) Exterior Gateway Protocol

Answer: C) Hybrid (Advanced Distance Vector) Protocol

66. Administrative Distance (AD) is used to:

A) Measure bandwidth

B) Rate the trustworthiness of a routing information source

C) Count hops

D) Encrypt data

Answer: B) Rate the trustworthiness of a routing information source

67. Which has a lower Administrative Distance (is more trusted)?

- A) RIP
- B) OSPF
- C) Connected Interface / Static Route
- D) BGP

Answer: C) Connected Interface / Static Route

68. Convergence in routing means:

- A) All routers have a consistent view of the network
- B) Routers are rebooting
- C) Data is being encrypted
- D) Packets are colliding

Answer: A) All routers have a consistent view of the network

69. In a Class A network, the default subnet mask is:

- A) 255.0.0.0
- B) 255.255.0.0
- C) 255.255.255.0
- D) 255.255.255.255

Answer: A) 255.0.0.0

70. How many valid host IPs are available in a /24 subnet?

- A) 256
- B) 255
- C) 254
- D) 250

Answer: C) 254 (256 minus Network ID and Broadcast IP)

71. A subnet mask of 255.255.255.192 creates how many subnets from a Class C network?

- A) 2
- B) 4

C) 8

D) 16

Answer: B) 4 (2 bits borrowed, $2^2 = 4$)

72. What is the broadcast address for the network 192.168.1.0/24?

A) 192.168.1.1

B) 192.168.1.254

C) 192.168.1.255

D) 192.168.1.0

Answer: C) 192.168.1.255

73. A "stub network" is a network that:

A) Has multiple connections to other networks

B) Has only one path to the external network

C) Is wireless

D) Is a ring topology

Answer: B) Has only one path to the external network

74. ICMP Echo Request and Echo Reply are used by:

A) Telnet

B) FTP

C) Ping

D) HTTP

Answer: C) Ping

75. Which is NOT a private IP range?

A) 10.0.0.0 - 10.255.255.255

B) 172.16.0.0 - 172.31.255.255

C) 192.168.0.0 - 192.168.255.255

D) 11.0.0.0 - 11.255.255.255

Answer: D) 11.0.0.0 - 11.255.255.255

76. The "Next Hop" in a routing table indicates:

- A) The final destination
- B) The next router to send the packet to
- C) The previous router
- D) The server address

Answer: B) The next router to send the packet to

77. An Exterior Gateway Protocol used on the Internet is:

- A) OSPF
- B) IS-IS
- C) BGP
- D) RIP

Answer: C) BGP

78. What does "Converged Network" mean in modern context?

- A) Combining Voice, Video, and Data on a single network
- B) A small network
- C) A network with one router
- D) An unstable network

Answer: A) Combining Voice, Video, and Data on a single network

79. MPLS stands for:

- A) Multi-Protocol Label Switching
- B) Multiple Path Link Service
- C) Main Path Linear System
- D) Multi-Point Link Standard

Answer: A) Multi-Protocol Label Switching

80. A Firewall operating at Layer 3 (Network Layer) filters based on:

- A) File names

- B) IP addresses
- C) MAC addresses
- D) Email subjects

Answer: B) IP addresses

81. A transparent bridge:

- A) Is invisible to the other devices
- B) Connects wireless only
- C) Does not filter
- D) Is made of glass

Answer: A) Is invisible to the other devices

82. Spanning Tree Protocol (STP) is used to:

- A) Prevent loops in a switched network
- B) Route packets
- C) Assign IP addresses
- D) Encrypt data

Answer: A) Prevent loops in a switched network

83. Which device creates multiple broadcast domains?

- A) Hub
- B) Unmanaged Switch
- C) Router
- D) Repeater

Answer: C) Router (Switches create collision domains, Routers create broadcast domains)

84. The "Intranet" typically uses:

- A) Proprietary protocols only
- B) TCP/IP protocols within an organization
- C) Only cables
- D) No protocols

Answer: B) TCP/IP protocols within an organization

85. A DMZ (Demilitarized Zone) in networking is:

- A) A highly secure internal zone
- B) A sub-network that exposes external-facing services to the Internet
- C) A zone for hackers
- D) A zone without computers

Answer: B) A sub-network that exposes external-facing services to the Internet

86. Which protocol is used to copy a file from one host to another?

- A) SMTP
- B) RCP (Remote Copy) / SCP
- C) SNMP
- D) UDP

Answer: B) RCP (Remote Copy) / SCP

87. Packet switching splits data into:

- A) Waves
- B) Analog signals
- C) Packets
- D) Circuits

Answer: C) Packets

88. Circuit Switching creates:

- A) A dedicated physical path for the duration of the connection
- B) Virtual paths
- C) Wireless signals
- D) Routing tables

Answer: A) A dedicated physical path for the duration of the connection

89. X.25 and Frame Relay are examples of:

- A) LAN technologies
- B) WAN packet-switching technologies
- C) Wireless standards
- D) Applications

Answer: B) WAN packet-switching technologies

90. In IPv6, the address size is:

- A) 32 bits
- B) 64 bits
- C) 128 bits
- D) 256 bits

Answer: C) 128 bits

91. Which is an advantage of IPv6 over IPv4?

- A) Larger address space
- B) Simpler header
- C) Built-in security
- D) All of the above

Answer: D) All of the above

92. The "3-way handshake" is a function of:

- A) IP
- B) TCP
- C) UDP
- D) ICMP

Answer: B) TCP

93. A "Gateway" is often implemented via:

- A) Software on a router or server
- B) A simple cable
- C) A repeater

D) A passive hub

Answer: A) Software on a router or server

94. Subnetting a Class C network with mask /30 provides how many usable host IPs per subnet?

A) 2

B) 4

C) 6

D) 1

Answer: A) 2 (Total 4, minus network and broadcast = 2)

95. Which organization manages IP address allocation globally?

A) IEEE

B) IANA

C) ISO

D) W3C

Answer: B) IANA (Internet Assigned Numbers Authority)

96. A collision domain is defined as:

A) The segment where a data collision can occur

B) The entire internet

C) A secure zone

D) A router interface

Answer: A) The segment where a data collision can occur

97. Switches break up _____ domains.

A) Broadcast

B) Collision

C) Routing

D) DNS

Answer: B) Collision

98. Routers break up _____ domains.

- A) Broadcast
- B) Collision
- C) Both Broadcast and Collision
- D) None

Answer: C) Both Broadcast and Collision

99. Which device performs "Packet Filtering"?

- A) Repeater
- B) Firewall / Router
- C) Hub
- D) Modem

Answer: B) Firewall / Router

100. In the OSI model, Internetworking takes place at layer:

- A) 1
- B) 2
- C) 3
- D) 4

Answer: C) 3 (Network Layer)